Xiao Zhang

Neuro-symbolic NLP / Computational Linguistics

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1npc.github.io | Google Scholar | Hugging Face | npcbank.org

Summary

PhD candidate in computational linguistics working on **neuro-symbolic NLP**: making neural language models produce, use, and respect symbolic structure — formal meaning representations, ontologies, and grammars. Hands-on across the model lifecycle, from pre-training and fine-tuning to GRPO post-training on multi-GPU clusters, and currently building **NPCBank**, an evaluation ecosystem for stateful game characters.

Education

University of Groningen 2022–2027
Ph.D. in Computational Linguistics. Thesis *Structure as Evidence*, submitted; defence February 2027. Supervisors: Johan Bos and Gosse Bouma.

Leiden University 2020–2022
M.Sc. in Computer Science and Artificial Intelligence. Thesis on news stance detection with syntactic structure; supervisor Suzan Verberne.

Xi'an Jiaotong University 2016–2020
B.Sc. in Information Engineering. Thesis on airway image analysis and data annotation.

Research Projects

Neural parsing into formal meaning CL 2025; IWCS 2025; ACM MM 2025
Taxonomy-enriched symbolic representations, retrieval-augmented LLM parsers, and vision–language parsing of tombstone images into structured meaning.

Certified constrained decoding ongoing
Hard, prefix-safe grammar constraints that guarantee well-formed DRS, AMR and knowledge-graph output under any tokenizer; only outputs that carry a validity certificate are released.

Symbolic knowledge in LLMs J. Web Semantics 2026; Findings of ACL 2026
Benchmarks for ontological reasoning and systematic knowledge coverage, evaluating 20+ open LLMs.

NPCBank (npcbank.org) under review, ICLR 2027
An LLM agent pipeline builds 20,655 evidence-grounded dialogue cases for 37 game NPCs across 21 capabilities, to test LLM role-play under changing state, on consumer GPUs, and over long interactions.

Industry Experience

WonderCV Jun.–Sep. 2021
Training pipelines for job-description classification (CNN-LSTM, BERT), Elasticsearch search and recommendation, and back-end services for client-facing NLP workflows.

Turbosight Jun.–Oct. 2020
Face-recognition and facial-expression models for embedded deployment; encryption routines for secure signal-data transmission.

Technical Skills

Neuro-symbolic	Formal meaning representations (DRS, AMR), ontologies and knowledge graphs (OWL, RDF, SPARQL, WordNet), constrained decoding.
Model training	Pre-training, full and LoRA fine-tuning, GRPO post-training; LLMs, VLMs, diffusion LMs.
Infrastructure	PyTorch, Hugging Face, vLLM, DeepSpeed, FSDP; SLURM GPU clusters (H100/H200).
Other	Python, Bash, Git, Linux, SQL, Elasticsearch. Mandarin Chinese (native), English (fluent).

Publications

Journal Articles

[J1] **Zhang, X.**; Bouma, G.; Bos, J. "Neural Semantic Parsing with Extremely Rich Symbolic Meaning Representations." *Computational Linguistics*, 51(1):235–274, 2025. **SJR Q1**

[J2] **Zhang, X.**; Lai, H.; Meng, Q.; Bos, J. "OntoURL: A Benchmark for Evaluating Large Language Models on Symbolic Ontological Understanding, Reasoning and Learning." *Journal of Web Semantics*, to appear, 2026. **SJR Q1**

[J3] Amin, M. S.; **Zhang, X.**; Anselma, L.; Mazzei, A.; Bos, J. "Semantic Processing for Urdu: Corpus Creation, Parsing, and Generation." *Language Resources and Evaluation*, 59:2469–2500, 2025. **SJR Q1**

Conference and Workshop Papers

[C1] Lai, H.; **Zhang, X.**; Nissim, M. "Multidimensional Consistency Improves Reasoning in Language Models." *Proceedings of AACL-IJCNLP*, to appear, 2026. **CORE B**

[C2] **Zhang, X.**; Meng, Q.; Chen, Y.; Wang, Y.; Bos, J. "KnowledgeBerg: Evaluating Systematic Knowledge Coverage and Compositional Reasoning in Large Language Models." *Findings of ACL*, 2026. **CORE A***

[C3] Meng, Q.; **Zhang, X.**; Ren, Z.; Visser, J. "When More Retrieval Hurts: Retrieval-Augmented Code Review Generation." *International Joint Conference on Neural Networks (IJCNN)*, 2026. **CORE B**

[C4] **Zhang, X.**; Bos, J. "Multi-Modal Semantic Parsing for the Interpretation of Tombstone Inscriptions." *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*, 6672–6681, 2025. **CORE A***

[C5] **Zhang, X.**; Meng, Q.; Bos, J. "Retrieval-Augmented Semantic Parsing: Improving Generalization with Lexical Knowledge." *Proceedings of the 16th International Conference on Computational Semantics (IWCS)*, 49–62, 2025.

[C6] **Zhang, X.**; Bos, J. "Is neural semantic parsing good at ellipsis resolution, or isn't it?" *Proceedings of the 16th International Conference on Computational Semantics (IWCS)*, 137–142, 2025.

[C7] **Zhang, X.**; Wang, C.; van Noord, R.; Bos, J. "Gaining More Insight into Neural Semantic Parsing with Challenging Benchmarks." *Fifth International Workshop on Designing Meaning Representations (DMR) @ LREC-COLING*, 162–175, 2024.

[C8] Meng, Q.; **Zhang, X.**; Ramackers, G.; Visser, J. "Combining Retrieval and Classification: Balancing Efficiency and Accuracy in Duplicate Bug Report Detection." *Proceedings of the Eighteenth International Conference on Software Engineering Advances (ICSEA)*, 75–84, 2023.

[C9] Wang, C.; **Zhang, X.**; Bos, J. "Discourse Representation Structure Parsing for Chinese." *Proceedings of the 4th Natural Logic Meets Machine Learning Workshop (NALOMA)*, 62–74, 2023.

Preprints

[P1] **Zhang, X.** "The Curse of Knowledge in Role-Playing Language Models." arXiv:2610.17327, 2026.

[P2] **Zhang, X.** "Structural Automata for Neural Semantic Parsing." arXiv:2610.13896, 2026.

Talks and Presentations

Poster, *KnowledgeBerg*, ACL 2026 (Findings), San Diego — Jul. 2026

Poster, *When More Retrieval Hurts*, IJCNN 2026, Maastricht — Jun. 2026

Invited talk, Utrecht University; earlier at Leiden University and Nanjing University — Jun. 2026

Oral, *Tombstone Inscriptions*, ACM Multimedia 2025, Dublin — Oct. 2025

Two orals, *Retrieval-Augmented Semantic Parsing* and *Ellipsis Resolution*, IWCS 2025, Düsseldorf — Sep. 2025

Poster, *Extremely Rich Symbolic Meaning Representations* (CL journal track), ACL 2025, Vienna — Jul. 2025

Oral, *Challenging Benchmarks*, DMR @ LREC-COLING 2024, Turin — May 2024

Teaching and Service

Guest lectures, Groningen: Semantic Web Technology, Computational Semantics, Responsible AI — 2025–2026

Programme committee, RAGE-KG 2026 @ ISWC — 2026

Reviewer: ACL Rolling Review (multiple cycles), ACM Multimedia, IEEE WCCI, IWCS — 2025–2026

Awards and Scholarships

China Scholarship Council PhD Fellowship — 2022–2026

XJTU-LIACS Scholarship (€14,000) — 2020, 2021

SiYuan Scholarship, Xi'an Jiaotong University — 2017